Codio Project File Edit Find View Tools Education Help Configure... Project Index (static) Configure... SANDIPAN-DEY

Guide

Collapse

Terminal

```
codio@kiwivatican-brownsunday:~/workspace$
```

Pixel Art Module 1

Setup

Our pixel art package depends on the `colorama` package, which allows you to change the foreground, background, and text colors in the terminal. The "pixels" are nothing more than two spaces (`' '`) with a background color. Using a virtual environment, install `colorama` in the terminal.

```
cd pixel_art/pixel_art
python3 -m venv pa-env
source pa-env/bin/activate
python3 -m pip install colorama
```

Start by importing from the `colorama` package. In particular, we want `init` to initialize our pixel art and `Back` which lets us set the background color. By setting `autoreset` to true, the colors of the terminal will return to normal after the pixel art image is displayed.

```
from colorama import init, Back, Style

init(autoreset=True)
```

Next, we need to create our "pixels". Colorama allows for only eight colors. We are going to use them all. Create a variable that corresponds to each color. Set its value to `Back` and the color name (the color name should be all caps) concatenated to a string with two spaces.

```
BLACK = Back.BLACK + '  '
RED = Back.RED + '  '
GREEN = Back.GREEN + '  '
YELLOW = Back.YELLOW + '  '
BLUE = Back.BLUE + '  '
MAGENTA = Back.MAGENTA + '  '
CYAN = Back.CYAN + '  '
WHITE = Back.WHITE + '  '
```

Users are going to pass a string representation of the color they want. So we need a way to convert the string into the Colorama objects we defined above. Use a dictionary that has strings for each color as the key and their Colorama object as the value.

```
COLORS_DICT = {
  'black' : BLACK,
  'red' : RED,
  'green' : GREEN,
  'yellow' : YELLOW,
  'blue' : BLUE,
  'magenta' : MAGENTA,
  'cyan' : CYAN,
  'white' : WHITE
}
```

Image Class

The `Image` class is the main vehicle for creating pixel art. Create a constructor that takes a width and height for the pixel art image. If no arguments are passed, default to 7 and 7. The `width` and `height` attributes are assigned their respective parameters. The `img` attribute takes its value from the `init_list` method.

```
class Image():
  def __init__(self, w=7, h=7):
    self.width = w
    self.height = h
    self.img = self.init_list()
```

The pixel art is nothing more than a two-dimensional list. Each element in the list will be a "pixel". In `init_list()`, create an empty list. Create a loop that runs as many times as the `height` attribute. Append an empty list to `lst` each time the loop runs. Return `lst` which becomes the initial value for the `img` attribute.

```
  def init_list(self):
    lst = []
    for row in range(self.height):
      lst.append([])
    return lst
```

Lab Question

What does it mean if a package is in the standard library?

- ○ The package resides on PyPI.
- ○ You don't have to use the `import` statement.
- ○ You have to use `pip` to install it.
- ● The package is included with Python.

The standard library refers to all packages included with Python. You do not have to use `pip` or any other tool to install them. Built-in packages do not reside on any external platform. You still have to use the `import` statement however.

Check It!

Next